

Podcast 28 — Cost Management

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Length target: 15 min **Repo references:** `domain-1-organizational/37-cost-explorer.md`, `38-cost-anomaly-detection.md`, `39-trusted-advisor.md`, `44-budgets.md`, `45-savings-plans.md`, `46-cost-and-usage-report.md`, `domain-2-new-solutions/03-compute-optimizer.md`

Topic & Scope

Cost Explorer, Budgets, Cost Anomaly Detection, Cost and Usage Report, Trusted Advisor, Compute Optimizer. Plus a Savings Plans recap. Cost questions weigh ~10% of SAP-C02.

Service Coverage Depth

Deep: - Cost Explorer — historical + forecast + RI/SP recommendations - Budgets — thresholds + Budget Actions (auto-remediation) - Cost Anomaly Detection — ML-based anomaly alerts - CUR — raw billing data - Trusted Advisor — recommendations across cost / perf / fault tolerance / security / service limits - Compute Optimizer — right-sizing for EC2 / ASG / EBS / Lambda / ECS Fargate

Brief: - Savings Plans recap (covered fully in podcast 13) - Cost Categories + Cost Allocation Tags

Structured Outline

1. **Open (30s)** — “Cost levers are: visibility, control, optimization. Six services do the heavy lifting.”
2. **Cost Explorer (2.5 min)** — interactive historical analysis, group by service/tag/account/region, forecast, SP/RI recommendations, Cost Optimization Hub
3. **Budgets (2.5 min)** — types (cost/usage/RI utilization/SP utilization/coverage), forecasted vs actual alerts, Budget Actions (SCP / IAM / stop EC2/RDS / EventBridge), Budgets Reports
4. **Cost Anomaly Detection (2 min)** — ML-based, monitors per service / linked account / cost category / tag, alerts via SNS / email
5. **Cost and Usage Report (2 min)** — most granular line-item data to S3, Athena/QuickSight integration, cost allocation tags must be activated
6. **Trusted Advisor (2 min)** — 5 categories: Cost / Performance / Fault Tolerance / Security / Service Limits, Business+ unlocks full checks
7. **Compute Optimizer (1.5 min)** — ML-based right-sizing for EC2 / ASG / EBS / Lambda / ECS on Fargate
8. **Savings Plans recap (1 min)** — Compute SP (EC2 + Fargate + Lambda) vs EC2 Instance SP vs SageMaker SP; 1y/3y; All/Partial/No Upfront
9. **Rapid-Fire Trap Drill (60s)**

Must-Mention Exam Tips

- Cost Explorer: 12-month historical + 12-month forecast, free
- Cost Explorer Resource-level granularity for last 14 days
- Cost Optimization Hub: aggregated savings recommendations
- Budgets: alerts at forecasted OR actual threshold; up to 5 alerts/budget
- Budget Actions: IAM policy / SCP / stop EC2 / stop RDS / EventBridge target
- First 2 budgets free; \$0.02 per budget per day after
- Cost Anomaly Detection: ML detects unusual spend; no threshold needed
- CUR: most detailed data; refresh up to 3×/day; CUR 2.0 via Data Exports
- Cost Allocation Tags must be ACTIVATED to appear in CUR + Cost Explorer reports
- Trusted Advisor 5 categories: Cost / Performance / Fault Tolerance / Security / Service Limits
- Trusted Advisor Basic + Developer plan: only 7 core checks; Business+ unlocks full set
- Compute Optimizer: needs 14 days of CloudWatch data for recommendations
- Compute Optimizer covers: EC2, ASG, EBS, Lambda (memory), ECS on Fargate, RDS (limited)
- Savings Plans: Compute SP (most flexible, EC2+Fargate+Lambda, up to 66%), EC2 Instance SP (one family in one Region, up to 72%), SageMaker SP
- RIs still needed for RDS, ElastiCache, Redshift, OpenSearch, DynamoDB (DynamoDB Reserved Capacity)

Must-Mention Exam Traps

- EXAM TRAP: Cost Explorer is RETROSPECTIVE; Budgets is PROACTIVE — different tools
- EXAM TRAP: Cost Anomaly Detection uses ML — no fixed threshold; complements Budgets, doesn't replace
- EXAM TRAP: Budget Actions can stop EC2 / RDS — but applies the role's permissions; scope tightly
- EXAM TRAP: Tags don't appear in CUR / Cost Explorer unless ACTIVATED in Billing
- EXAM TRAP: Historical tag data is lost if you activate tag late — only forward-looking
- EXAM TRAP: CUR is generated FROM management account — linked accounts see summary
- EXAM TRAP: Trusted Advisor full checks need Business+ plan
- EXAM TRAP: Trusted Advisor Service Limits checks — but Service Quotas is more precise
- EXAM TRAP: Compute Optimizer needs 14 days of metrics — new workloads not eligible
- EXAM TRAP: Savings Plans do NOT cover RDS / ElastiCache / Redshift — must use RIs
- EXAM TRAP: SP and RI sharing across Org is ON by default
- EXAM TRAP: Compute SP covers Lambda — easy to forget
- EXAM TRAP: Free tier covers small Compute Optimizer / Cost Explorer / Budgets use
- EXAM TRAP: Athena query on CUR scans whole file without partition pruning — costly

Key Decision Matrix

Need	Pick
Interactive historical cost analysis + forecast	Cost Explorer
Threshold alert + auto-action	Budgets
ML-based unusual spend detection	Cost Anomaly Detection
Custom dashboards on raw billing data	CUR → Athena → QuickSight
Best-practice recommendations (cost + perf + security)	Trusted Advisor
Right-size EC2 / Lambda / EBS	Compute Optimizer
Discount for EC2 + Fargate + Lambda	Compute Savings Plan
Discount for one EC2 family steady state	EC2 Instance Savings Plan
Discount for RDS / ElastiCache / Redshift	Reserved Instances
Discount for DynamoDB	DynamoDB Reserved Capacity
Track cost per team / product	Cost Allocation Tags + Cost Categories
Auto-stop EC2 when budget breached	Budgets + Budget Action

Tone & Style

- Frame as “visibility → control → optimize”
- Repeat “Cost Explorer retrospective, Budgets proactive”
- For Savings Plans, recap one-liner: “Compute SP for new modern workloads, EC2 Instance SP for locked-in”

Rapid-Fire Closer

“Threshold alert?” — “Budgets.” “ML anomaly?” — “Cost Anomaly Detection.” “Raw billing data?” — “CUR.” “Right-size EC2?” — “Compute Optimizer.” “Best-practice checks?” — “Trusted Advisor.” “RDS discount?” — “RI (NOT Savings Plan).”